

AI Yoga Posture Corrector with Real-Time Voice and Text Feedback for Online Fitness Training

Objective: Developed an AI-powered system capable of recognizing and correcting yoga poses in real time using pose estimation techniques. The system provided immediate corrective feedback through both text and voice to enhance remote fitness training experiences.

Literature Review Table:

<u>S.No</u>	Title of the Paper and Year	Algorithm Used	Dataset Used (<i>paper wording</i>)	Performance Measures	Future Work / Gap Identified (<i>short</i>)	PDF Link
1	PosePilot: An Edge-AI Solution for Posture Correction in Physical Exercises (2025)	CNN-LSTM, BiLSTM with Multi-head Attention	In-house video dataset: 6 asanas, 14 participants, 336 videos (4 angles per pose); keypoints by	Real-time posture correction; optimized for edge-device deployment T High recognition accuracy (avg. 97.52%) and very low correction error (avg. 0.00138 MSE) across all poses.	Further optimise end-to-end latency for edge devices; evaluate beyond Yoga (physiotherapy,	Open PDF Open PDF view pdf

			MediaPipe; angles as features.		sports); develop explainable pose- correction.	
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2	Yoga Pose Recognition and Motion Analysis for Home-Based Fitness (2025)	OpenPose + Scoring System	Personal training video dataset	Static poses achieved 99.9% accuracy , dynamic poses 99.0% , with improved Warrior Pose detection through refined leg position rules.	Lacks integrated corrective audio/text feedback	Open PDF Open PDF view link
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3	DeepYoga: A Real-Time Yoga Pose Recognition System (2024)	CNN	Yoga Poses Dataset: 5 classes (Tree, DOWNDOG, Goddess, Plank, Warrior2); 1,551 initial images; post-processing ~1,817 train / 428 test; landmarks via MediaPipe.	Pose recognition accuracy: 99.02%	Expand dataset (more poses, diverse body types); apply transfer learning; integrate into web/mobile apps.	Open PDF view pdf
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4	Deep Learning-Based Yoga Posture Recognition Using Y_PN-MSSD (2023)	TFLite Movenet, MobileNet SSD	Yoga pose image dataset	Proposed Y_PN-MSSD model outperforms PoseNet CNN with lower FPR (0.01), higher precision (0.98), recall (0.99), and accuracy (0.99), achieving the highest overall accuracy of 99.88% among compared methods.	Needs integration with correction and communication features	Open PDF view pdf
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5	YPose: Efficient CNN Model for Yoga Pose Recognition (2023)	EfficientNet + Dense layers	Yoga-82 benchmark dataset: 82 classes, ~28.4k images; 3-level hierarchy.	Accuracy: 93.3% (static pose classification)	Apply more augmentation; explore improved segmentation; extend to deeper backbones.	Open PDF
6	Ensemble Learning with Multi-Head Attention for Yoga Pose Recognition (2024)	ResNet, VGG with multi-head attention mechanism	Custom seven-pose dataset; supplementary materials & data available in article/code archive.	Accuracy: 93.8%, inference speed: 180–300 fps	Increase sample size; explore DenseNet/Transformer; improve accuracy & efficiency.	Open PDF

7	A Yoga Pose Difficulty Level Estimation Method Using OpenPose for Self-Practice System to Yoga Beginners (2024)	OpenPose BODY25 Model; area, centroid, slope calculation s	~100+ yoga poses collected from Internet + instructor images; 12 BODY25 keypoints per pose; classified into five difficulty levels	Yoga poses are classified into five difficulty levels from Beginner (Lv.1) to Master (Lv.5) . Examples show difficulty scores ranging from 25.51 (Beginner) to 101.45 (Master).	Incorporate body flexibility; expand dataset; enhance real-time feedback & usability for diverse age groups	Open PDF Open PDF
8	Real-time Pose Correction and Wellness Tracking System for Enhanced Yoga Practice (2025)	PoseNet + KNN; Mediapipe for cardio; computer vision for meditation module	Custom dataset : 1578 images (4 yoga poses: Goddess, Mountain, Plank, Garland) from various sources incl.	Pose accuracy (Mountain 80%, Goddess 86%, Garland 70%, Plank 72%); cardio 82%; meditation 92%	Increase pose set diversity; refine correction algorithms; add social features; enable live feedback with instructors	Open PDF Open PDF

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9	Pose Estimation and Correcting Exercise Posture (2021)	OpenPose; CNN; DTW + k-NN classifier	Exercise videos (Bicep Curl, Front Raise, Shoulder Shrug, Shoulder Press); uses OpenPose 18-point model output	Feedback on form using geometry & ML; detects errors in joint angles and motion patterns	Extend to more exercises; mobile/web app; targeted suggestions with graphical simulation	Open PDF Open PDF
10	Estimation of Yoga Postures Using Machine Learning Techniques (2022)	EpipolarPose, OpenPose, PoseNet, MediaPipe	~6000 images of 5 poses (Tree, Triangle, Half-moon, Mountain, Warrior) from S-VYASA dataset (75% train / 25% test)	MediaPipe highest accuracy (78–91%); PoseNet ~80%; OpenPose ~70%; EpipolarPose ~50%	Extend to advanced poses; integrate correction feedback; optimize single-camera performance	Open PDF Open PDF

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11	A View Independent Classification Framework for Yoga Postures (2022)	AlphaPose pose estimation (136 keypoints) + Random Forest classifier	In-house yoga video dataset: 51 subjects, 20 asanas, 4 camera angles (~72k frames)	Frame-wise F1: 99.7%; Subject-wise F1: 97.99%; Camera-wise tests show robustness	Release dataset/code; improve unseen-view accuracy; address occlusion/complex poses	Open PDF
12	3DYoga90: A Hierarchical Video Dataset for Yoga Pose Understanding (2023)	BlazePose 3D keypoint extraction + DNN classifiers (Small/Base/Large)	3DYoga90 dataset: 6,177 RGB videos & 5,526 3D skeleton sequences; 90 fine-grained poses in 3-level hierarchy	L1 acc: 95%; L2 acc: 91%; L3 acc: 86% (DNN-Base best)	Extend to RGB video recognition; use graph neural networks; expand dataset with more poses/sources	Open PDF
13	REAL-TIME Recognition of Yoga Poses Using Computer Vision for Smart Health Care (2022)	tf-pose-estimation for body; MediaPipe Hands; joint-angle features; XGBoost classifier	YOGI dataset: 10 yoga postures (400–900 images each) & 5 mudras (500 each); skeleton features from pose & hand models	Yoga pose acc: ~98%; Mudra acc: 99.2% (XGBoost)	Expand dataset; add audio-feedback; integrate IoT for multi-user health monitoring	Open PDF
14	Yoga Pose Recognition using Deep Learning (Year per paper)	MediaPipe keypoint extraction → CNN+LSTM classification → cosine similarity–based correction.	Six poses, webcam capture, keypoints via MediaPipe in JSON format.	The model achieved high accuracy with a training accuracy of 99.49% and a testing accuracy of 99.70%.	Expand to more poses, improve overlapping-body detection, make portable real-time device, apply in sports/healthcare.	Open PDF

Theory Part for Research Papers

R. Gadhvi,[1] et al presented PosePilot, a novel Edge-AI system that integrated pose recognition with real-time personalized corrective feedback, particularly demonstrated on yoga. The method used a Vanilla LSTM for temporal context and a BiLSTM with multi-head Attention to focus on key limb angles, enabling accurate error detection while remaining lightweight and deployable on edge devices. PosePilot introduced a high-quality video dataset and ensured continuous feedback during movements.

S.-C. Yeh,[2] et al proposed a home-based system combining static yoga poses and dynamic exercises, using OpenPose for precise keypoint detection and real-time feedback. This system included a scoring mechanism to quantify pose accuracy, achieving a notable 99.9% prediction accuracy, aimed especially at improving accessibility and safety in home fitness routines.

A.Sharma,[3] et al developed a self-guided yoga pose identification system with correction features, including hand mudra recognition using the YOGI dataset (with 10 poses and 5 mudras). Skeleton-based feature extraction and joint angle computation fed into machine learning models, where XGBoost with randomized search (RandomSearch CV) achieved 99.2% accuracy.

S. Kim,[4] et al created the 3DYoga90 dataset, expanding pose categories from 82 to 90, and providing both RGB yoga videos and corresponding 3D skeleton sequences, supported by a hierarchical label taxonomy. This rich resource enhanced possibilities for developing advanced yoga action recognition and pose assessment models.

D. M. Kishore,[5] et al explored the recognition of five classic yoga poses (Ardha Chandrasana, Tadasana, Trikonasana, Veerabhadrasana, Vrukshasana), comparing four deep-learning-based pose estimation models: EpipolarPose, OpenPose, PoseNet, and MediaPipe. Their results found MediaPipe provided the highest accuracy on a dataset sourced from S-VYASA University.

I.Chaudhary,[6] et al designed an AI-based personalized yoga trainer using OpenCV for capture and MediaPipe (with 33 body landmarks) for pose detection. The system assessed body landmarks and joint angles in real time and offered corrective feedback to improve alignment and safety during yoga practice.

S. Nagargoje,[7] et al work used OpenPose to generate 3D skeletal body representations, which were then processed through a linear regression model to classify yoga poses in real time. It offered straightforward deployment for studios or home environments and allowed long-term performance tracking.

A.Sharma,[8] et al crafted a system to analyze Surya Namaskar sequences using MediaPipe and CNN-based deep learning models, classifying key poses like Ashtanga Namaskar and Bhujangasana, achieving approximately 98.68% accuracy. This enabled practitioners to confirm correctness of their sequences in real time.

P. Choudhary,[9] et al review analyzed 52 papers on yoga pose detection and feedback, highlighting methods such as CNNs, RNNs, skeleton-based, and pose graph techniques, as well as feedback mechanisms including virtual assistants and haptics. The authors outlined current limitations in datasets, real-time feedback, and generalizability, and proposed deep learning-based solutions to improve accuracy, efficiency, and personalization.

S. K. Yadav,[10] et al introduced YPose, a streamlined deep CNN tailored for real-time yoga pose recognition from single RGB images. The pipeline segmented the Region of Interest (ROI), extracted features using an EfficientNet-based backbone, refined with densely connected layers, and classified via global average pooling. Evaluated on the Yoga-82 dataset, YPose achieved a state-of-the-art accuracy of 93.28%, a substantial improvement over the previous benchmark (~79.35%).

Mustafa Chasmai,[11] et al proposed a novel ensemble-learning framework incorporating multi-head attention, merging features extracted from ResNet101 and VGG19 via Mixup-augmented training. This model delivered robust results with 100% training accuracy, 89.94% validation accuracy, and 93.79% test accuracy at a fast inference speed of 297 frames per second, showcasing both performance and practicality for real-time fitness applications.

A.L. Shih,[12] et al presented an objective difficulty-level estimation system to support beginner self-practice in yoga, leveraging six upper/lower-body keypoints extracted by OpenPose (BODY25). The method calculated weighted centers of mass, centroid line slopes, ground-contact points, and posture support types to assign each yoga pose into one of five difficulty levels. The results were validated against subjective ratings from one instructor and ten users, proving reliable and integrated into embedded systems.

S. Garg,[13] et al YogaConvo2d, a custom CNN trained on MediaPipe-processed skeleton images, they achieved 99.62% validation accuracy, outperforming models like VGG16, InceptionV3, NASNetMobile, and InceptionResNetV2. Designed for low-latency, real-world settings, this model was ideal for home-based yoga guidance systems.

P. Kulkarni,[14] et al combined CNN with LSTM to monitor yoga poses in video feeds. CNN extracted keypoint-based features while LSTM captured temporal transitions to judge pose correctness. The system classified poses as correct or incorrect and provided real-time feedback via text or speech, bridging deep learning and practical wellness tracking.

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